Problem B

Predicting a Penalty Kick's Goal-Line Crossing from Two Early Positions

Abstract

How precisely can two positions, observed 50 and 100 ms after a penalty kick, locate the ball at the goal line? We weight trajectories from three aerodynamic models by their agreement with the readings and estimate the crossing location conditional on direct entry. For the official example, with coordinates rounded to the nearest centimetre and horizontal wind bounded at 3 m/s, the most probable cell is centred 2.562 m from the goal centre on the positive-y side and 2.074 m above ground. Six cells spanning 0.3572 m^2 contain 90.52% of the conditional probability. Two admissible trajectories reproduce the displayed positions to numerical precision yet cross 8.76 cm apart. Among 48 held-out simulated kicks, the two-frame forecast had a median location error of 0.133 m for the 33 direct entries. Its nominal 90% region contained 30 of those crossings, compared with all 33 for the 100 ms position alone. With eight times as many trajectory proposals, the two-frame configuration reduced mean region area by 2.722 m^2 and the direct-entry Brier score by 0.0556 relative to the single-frame forecast. In three professional match penalties, positions sampled 40 and 80 ms after the first live frame violated the models' acceleration bounds under the assumed launch conditions and coordinate accuracy. The later frame alone yielded one forecast, 0.849 m from the tracking reference. Field validation therefore requires independent release timing and a calibrated tracking-error model.

1 Flight geometry in the first 100 milliseconds

By the second observation, the ball in the official example has travelled 2.96 m toward a goal plane 11 m from the penalty spot. The two readings determine its mean velocity over 50 ms. The remaining 8.04 m make the goal crossing sensitive to forces that the early positions constrain only partially. The contest asks for the most likely point of entry, the uncertainty in both coordinates and a test of the forecast [1].

1.1 Release geometry and first-event outcomes

Take x toward the goal, y across its face from the centre, and z upward from the ground. The ball centre starts at $\boldsymbol{r}_0 = (0, 0, 0.11)$ m. The prescribed positions, in metres, are

$$\boldsymbol{r}(0.050\,\mathrm{s}) = (1.50, 0.45, 0.44), \qquad \boldsymbol{r}(0.100\,\mathrm{s}) = (2.96, 0.86, 0.74). \tag{1}$$

We take the release position and times as exact. Each later coordinate q, displayed to the nearest centimetre, admits the half-open interval $[q - \epsilon, q + \epsilon)$, where $\epsilon = 0.005$ m. Further tracking error and uncertainty in release time would require a separate measurement model.

We follow the ball until its centre crosses $x = 11$ m or descends to one ball radius above ground. A crossing through the opening, $-3.66 < Y < 3.66$ m and $0 < Z < 2.44$ m, is a direct entry G. This ball-centre criterion neglects collisions with the frame, rebounds and goalkeeper interventions. The other outcomes are a crossing outside the opening, earlier ground contact, or neither event by 1.25 s. We estimate all four probabilities and the joint distribution of $Q = (Y, Z)$ conditional on G.

1.2 A lower bound on wind speed under drag alone

The mean horizontal velocities in the two successive intervals are $(30.0, 9.0)$ and $(29.2, 8.2)$ m/s. With drag alone and constant horizontal wind $\boldsymbol{w}$, the horizontal heading in the air frame remains fixed. The displacements $\boldsymbol{p}_i$ from release must therefore satisfy

$$\det(\boldsymbol{p}_1 - t_1\boldsymbol{w},\, \boldsymbol{p}_2 - t_2\boldsymbol{w}) = 0, \qquad \|\boldsymbol{w}\| \geq \frac{|\det(\boldsymbol{p}_1, \boldsymbol{p}_2)|}{\|t_2\boldsymbol{p}_1 - t_1\boldsymbol{p}_2\|}. \tag{2}$$

The determinant is the signed area spanned by two planar vectors. The displayed coordinates require at least 14.849 m/s of wind. At fixed wind, the determinant is linear in each coordinate separately and attains its extrema over the closed rounding boxes at their corners. Across the half-open rounding boxes, 4.353 m/s is the infimum wind speed compatible with horizontal collinearity. Our 3 m/s wind cap therefore rules out a drag-only explanation. The geometry leaves the source of the lateral force unresolved.

In still air, each position bounds the slope of a line through the origin. For these positive coordinates, the closure of its admissible interval is

$$\left[\frac{y_i - \epsilon}{x_i + \epsilon},\, \frac{y_i + \epsilon}{x_i - \epsilon}\right].$$

The two intervals' closures first touch at $\epsilon = 7.279$ mm. Thus 5 mm precision resolves a departure from a straight horizontal path in this example. A 1 cm bound admits a straight path.

2 Aerodynamic models and conditional inference

2.1 Force laws and parameter ranges

Let $\boldsymbol{v} = \dot{\boldsymbol{r}}$ be the ground velocity, $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{w}$ the air-relative velocity, and $U = \|\boldsymbol{u}\|$. We propagate three force models,

$$\begin{aligned} \boldsymbol{a}_{\mathrm{D}} &= \boldsymbol{g} - \kappa U \boldsymbol{u}, \\ \boldsymbol{a}_{\mathrm{S}} &= \boldsymbol{a}_{\mathrm{D}} + \boldsymbol{\beta} \times \boldsymbol{u}, \\ \boldsymbol{a}_{\mathrm{E}} &= \boldsymbol{g} - K C_D U \boldsymbol{u} + K C_L U^2 \boldsymbol{n}. \end{aligned} \tag{3}$$

D represents quadratic drag. S adds a transverse force with constant coefficient vector $\boldsymbol{\beta}$. In E, $K = \rho\pi R^2/(2m)$ and $\boldsymbol{n}$ points along $\boldsymbol{\omega} \times \boldsymbol{u}$. Lift vanishes when this cross product is zero. We use $g = 9.81\ \mathrm{m/s^2}$ downward, $R = 0.11$ m, $m = 0.43$ kg and $\rho = 1.20\ \mathrm{kg/m^3}$. With all inputs in SI units, E uses

$$\begin{aligned} C_D &= 0.155 + \frac{0.346}{1 + \exp[(U_c - 12.19)/1.309]} + \delta, \qquad U_c = \min(29.9, \max(9.3, U)), \\ C_L &= 0.38 \tanh\Big(5R\|\boldsymbol{\omega} \times \boldsymbol{u}\|/U^2\Big). \end{aligned} \tag{4}$$

The drag transition comes from a wind-tunnel fit for a nonspinning Teamgeist ball [2]. We clip its speed input to the initial-speed range of a separate 29-flight study [3]. The offset δ and the saturating lift law are modelling assumptions.

The prior probabilities of D, S and E are $0.2, 0.4, 0.4$. We draw constant horizontal wind uniformly from a disk of radius 3 m/s. Independent uniform launch-velocity ranges are $[18, 38]$, $[-12, 12]$ and $[1.5, 11]$ m/s along x, y, z. In D and S, κ is uniform on $[0.006, 0.024]\ \mathrm{m^{-1}}$. In S, each component of $\boldsymbol{\beta}$ is uniform within $\pm 0.8\ \mathrm{s^{-1}}$. In E, each component of $\boldsymbol{\omega}$ is uniform within ± 65 rad/s and δ is uniform on $[-0.04, 0.27]$.

2.2 Trajectory weighting and goal-plane probabilities

After drawing wind and aerodynamic parameters, we centre a launch-velocity proposal on a gravity-based estimate corrected three times against the latest visible position. We draw velocities from a mixture of this local uniform proposal (half-width $0.007\,\mathrm{m}/t$ per axis at observation time t) and the original prior (2%). A trajectory inside every observed interval receives the prior-to-proposal density ratio. Others receive zero weight. The mean weight over all proposals estimates each model's evidence. Multiplying by the model priors and normalizing yields the posterior weights. For propagation to the goal, systematic resampling caps each model at 512 trajectories per test kick or 8,192 per seed for the official case. The official runs stayed below the cap.

We integrate by fourth-order Runge–Kutta with steps of 25 ms through the official observations and at most 12.5 ms thereafter. Linear interpolation locates and orders plane and ground events. Across four numerical checks, including independent DOP853 integrations in SciPy [4], discrepancies stayed below 0.171 mm in position and 0.038 ms in time. After summing weights by first event, we distribute the direct-entry mass across a 30×10 equal-area grid. The centre of the highest-mass cell is the point forecast. Cells ranked by decreasing mass form the raw joint region once their cumulative mass reaches 90%. Zero total weight leaves the forecast undefined. Zero direct-entry weight leaves its conditional map undefined.

3 Goal-line forecast for the official kick

3.1 Modal cell and joint prediction region

The highest-probability cell is centred at $(Y, Z) = (2.562, 2.074)$ m and holds 29.14% of the probability conditional on direct entry (Fig. 1). Its centre is 1.098 m in from the positive-y post and 0.366 m below the crossbar. The six most probable cells form the rectangle

$$2.196 \le Y \le 2.928 \text{ m}, \qquad 1.708 \le Z \le 2.196 \text{ m}. \tag{5}$$

Together they cover 0.3572 m^2 and contain 90.52% of the conditional probability. Equation (5) gives the edges of a joint two-dimensional region. The median time from release to crossing is 0.399 s, with 5th and 95th percentiles of 0.392 and 0.407 s.

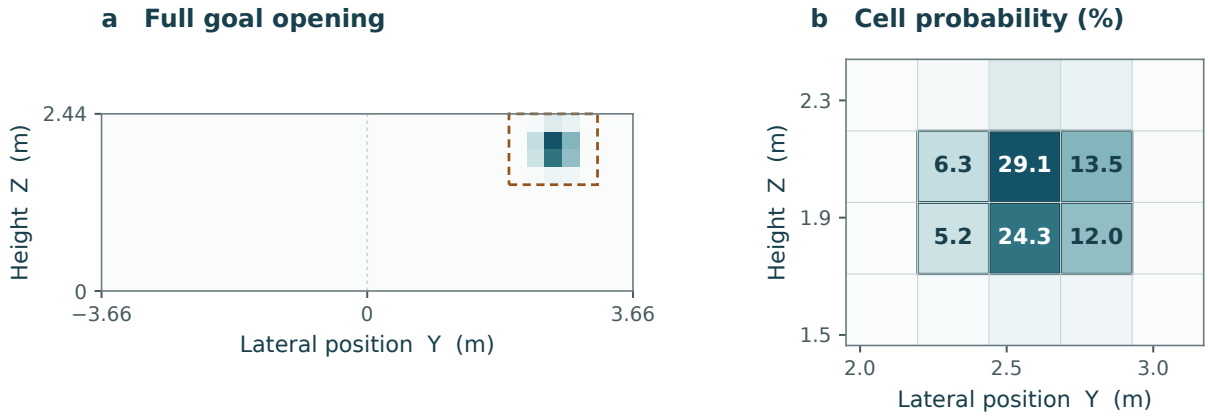


Figure 1. Goal-plane location at $x = 11$ m for the official kick. The left panel locates the enlarged area within the goal. The right panel gives the conditional probability in each outlined square cell (side length 0.244 m) of the raw 90% region. Darker shading indicates greater mass throughout.

We combined two independent runs of 131,072 proposals per model per seed. All 3,469 positive-weight trajectories crossed inside the opening, giving an estimated direct-entry probability of 1 in this sample. The posterior model weights were $0, 0.346, 0.654$ for D, S and E. The zero weight for D agrees with the geometric bound in Eq. (2). The official example supplies no measured crossing for comparison.

3.2 Sensitivity to grid choice and force law

Both seeds selected the same cell, as did mixtures in which E accounted for 10% to 90% of the combined S/E prior. A finer grid moved the mode to $(2.501, 2.013)$ m, and shifting the original grid by half a cell moved it to $(2.684, 1.952)$ m. The millimetre digits in the reported mode specify a grid centre. The six-cell region gives the scale of its positional uncertainty.

Kinematic fits illustrate the effect of force assumptions over the remaining eight metres. A least-squares fit of launch velocity under gravity alone gives $(3.217, 1.957)$ m. Constant acceleration fitted exactly to release and both frames gives $(2.495, 1.920)$ m. This spread measures sensitivity to the extrapolation law.

4 A lower bound on two-position prediction error

4.1 Indistinguishable readings and distinct crossings

The official readings at 50 and 100 ms admit two physically distinct crossings even when we fit the displayed coordinates to numerical precision. In model S we set the wind to zero and fixed β_x at either -0.5 or $+0.5$ s^{-1}. For each choice we solved the six early-coordinate equations for the three launch-velocity components and κ, β_y, β_z. Both solutions lie within the prior ranges. Their largest coordinate residual was below 5×10^{-13} m, and an independent integration with MATLAB `ode45` reproduced both crossings [5].

The two fitted drag coefficients agree to the displayed precision, 0.02007 m^{-1}. The corresponding (β_y, β_z) pairs are $(-0.206, -0.463)$ and $(0.085, -0.251)$ s^{-1}. At any instant, the component of $\boldsymbol{\beta}$ parallel to $\boldsymbol{u}$ contributes no transverse force. As gravity changes the direction of $\boldsymbol{u}$, that component can begin to affect the flight. Small compensating changes to launch velocity allow the two force histories to match both early readings.

The crossings are $Q_A = (2.57378, 2.00251)$ m and $Q_B = (2.62831, 1.93396)$ m, separated by 8.76 cm. Both trajectories give exactly the same displayed centimetre readings. A point predictor receiving those readings must return the same $\widehat{Q}$ for both. By the triangle inequality, its error on at least one member of this pair satisfies

$$\max\{\|\widehat{Q} - Q_A\|, \|\widehat{Q} - Q_B\|\} \;\geq\; \tfrac{1}{2}\|Q_A - Q_B\| \;\approx\; 0.0438 \text{ m}. \tag{6}$$

This lower bound applies whenever both flights are admissible. It requires no assumption about their relative frequency.

4.2 Distinguishing the flights with a third observation

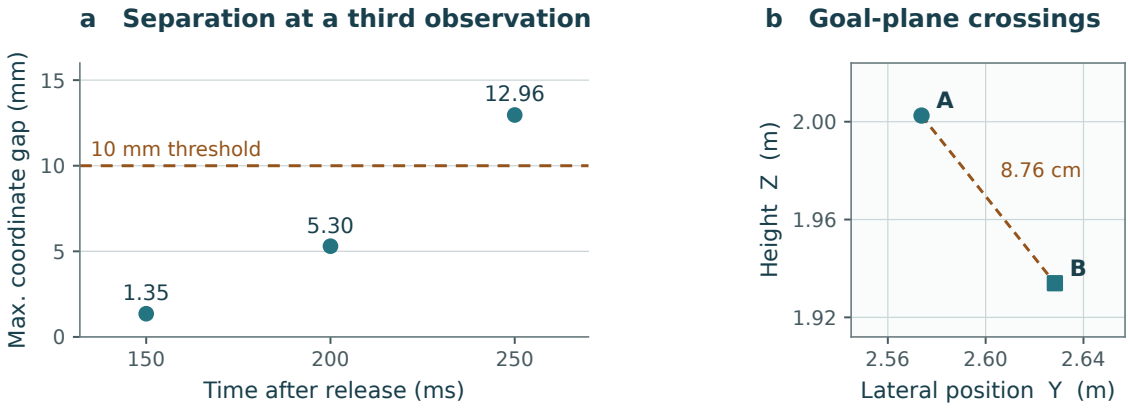


Figure 2. Separation of two trajectories that reproduce both official readings. Left, the largest difference along any measured axis at each candidate third observation. The dashed line marks twice the 5 mm coordinate-error bound. Right, the two goal-plane crossings on equal spatial scales.

At 150, 200 and 250 ms the largest coordinate differences are 1.35, 5.30 and 12.96 mm, respectively (Fig. 2). With deterministic centimetre rounding, the heights already display as 1.26 and 1.25 m at 200 ms. Under a general ± 5 mm error bound, either flight can still yield the same reading at that time. Of the three times examined, only 250 ms separates this pair for every admissible error.

5 Forecast performance on held-out simulated kicks

5.1 Test design and empirical coverage

A separate adaptive solver generated test trajectories with a different speed-dependent drag law. The generator included spin-free flight, stable spin with decaying lift, and low-spin flight with a time-varying wake. Slowly spinning balls can experience unsteady forces [6]. We prescribed the wake amplitude and period in the generator and omitted the wake from the forecasting models. We randomly ordered one kick per family within each block of three and split whole kicks into development (9), calibration (36) and blind test (48). At a common 100 ms decision time, we supplied the release position alone, release and either frame, or release and both frames. We froze forecasts before scoring the endpoints.

Table 1. Blind forecasts at a common 100 ms decision time. Location errors, hits and mean region areas use the same 33 direct entries. A hit places the true crossing in a selected cell. The Brier score averages $(P(G) - \mathbf{1}_G)^2$ over all 48 kicks, including ten outside crossings and five earlier ground contacts.

	Point error (m)			Raw 90% region		Calibrated region	
Input positions	Median	Mean	Brier	Hits	Area (m^2)	Hits	Area (m^2)
Release only	2.347	2.404	0.2898	23/33	11.431	33/33	17.861
Release + 50 ms	0.127	0.259	0.1153	32/33	3.704	28/33	2.087
Release + 100 ms	0.131	0.216	0.1057	33/33	3.022	27/33	1.290
Release + both	0.133	0.181	0.0501	30/33	0.299	29/33	0.220

For each of the 19 direct entries in the calibration set, we ranked cells by decreasing mass and scored the entry by the cumulative mass preceding its true cell. For each input choice, the cutoff was the $\lceil 0.90(19+1) \rceil = 18$th ordered score. The calibrated region includes cells whose preceding mass is at most this cutoff. With both frames, the cutoff was 0.8104. The calibrated region contained 29/33 test entries with mean area 0.220 m^2. The raw region contained 30/33, with a descriptive 95% Wilson interval of 76.4–96.9% for coverage [7].

5.2 Paired comparisons and kinematic baselines

Relative to the late frame alone, the two-frame configuration reduced mean raw region area by 2.722 m^2 and the Brier score by 0.0556 (Table 1). The smaller raw regions covered 30 of 33 crossings, down from 33 of 33. For the signed differences (both minus late), 95% intervals from 2,000 whole-kick bootstrap resamples were $[-2.979, -2.466]$ m^2 and $[-0.0988, -0.0086]$, respectively. The mean point-error difference was -0.034 m, with interval $[-0.082, 0.010]$ m. Each model used 16,384 proposals with both frames and 2,048 with one, so the comparison combines extra information with greater sampling effort.

For the nine stable-spin entries, mean point error fell from 0.492 to 0.311 m with the added frame. For the eight wake entries it rose from 0.131 to 0.180 m. Gravity alone had a lower median error (0.104 m) across all 33 entries than the two-frame aerodynamic forecast, but a higher mean (0.203 m) and Brier score (0.125). The constant-acceleration fit gave a mean error of 0.258 m for 32 entries and a Brier score of 0.146 over all 48 kicks.

6 Compatibility with 25 Hz match tracking

6.1 Match timing and reference crossings

The seven matches in the IDSSE v1 elite-soccer dataset contained three executed penalties, all recorded as goals [8]. We translated the longitudinal coordinate by 41.50 m into the penalty-spot frame and took the first live frame after a dead-ball interval as an exact anchor. The next two tracking frames fell at +40 and +80 ms. Both input choices were evaluated at +80 ms with the original priors and 5 mm coordinate bound. We interpolated each reference crossing between later frames from the same system, at least 0.32 s after the forecast cutoff. We isolated inputs by time and locked predictions before scoring. Prior knowledge of these crossings from data screening makes the comparison retrospective.

Table 2. Three penalties observed on the match clock. Each row gives the tracked goal-plane crossing, the first displacement or second difference constraining the fit, and the minimum acceleration magnitude it requires under the assumed launch conditions. Here $\Delta^2 x = x_2 - 2x_1 + x_0$.

Fixture	Reference (Y, Z) (m)	Observed quantity (m)	Min. accel. (m/s^2)
Cologne–Bayern	$(2.750, 0.160)$	$z_1 - z_0 = -0.030$	106.25
Düsseldorf–Regensburg	$(2.919, 1.179)$	$\Delta^2 x = 0.510$	309.38
Düsseldorf–Kaiserslautern	$(3.070, 0.466)$	$\Delta^2 x = 1.100$	678.13

6.2 An acceleration bound for consecutive positions

In constant wind, drag dissipates air-relative kinetic energy and transverse acceleration does no work in the air frame. Hence $\dot{U} \le g$ and $U \le \sqrt{38^2 + 12^2 + 11^2} + 3 + 9.81(0.080) = 45.125$ m/s over the first 80 ms. In Eq. (3), drag and lift are orthogonal. Bounding their resultant in quadrature and adding g gives upper acceleration bounds of 58.681, 89.170 and 102.648 m/s^2 for D, S and E. For E we used $C_D \le 0.771$ and $C_L \le 0.38$. Thus $A_* = 102.648$ m/s^2 bounds all three models.

For equally spaced coordinates q_0, q_1, q_2, the true second difference integrates acceleration against a triangular kernel of area h^2, with $h = 0.040$ s. An exact anchor and errors bounded by ϵ in the next two frames require

$$|q_2 - 2q_1 + q_0| \le A_* h^2 + 3\epsilon, \qquad z_1 - z_0 \ge v_{z,\min} h - \tfrac{1}{2} A_* h^2 - \epsilon. \tag{7}$$

The vertical condition uses the launch prior $v_{z,\min} = 1.5$ m/s. Regensburg's successive x-increments of 0.71 and 1.22 m require at least $(0.510 - 0.015)/h^2 = 309.38$ m/s^2. The Kaiserslautern record repeats its anchor, then advances 1.10 m. Cologne violates the vertical condition by only 2.9 mm, so its rejection is sensitive to the launch assumption. In each case the required acceleration exceeds A_* (Table 2). These inequalities exclude every two-frame fit under the stated assumptions.

6.3 Forecast from the 80 ms position alone

With only the +80 ms frame, neither seed produced positive weight for Cologne or Kaiserslautern. Both yielded a forecast for Regensburg. Its modal cell centre, $(2.074, 1.098)$ m, missed the reference by 0.849 m. The raw region covered 4.941 m^2 and contained the reference. A second seed selected the same centre with area 4.584 m^2. With the same late frame, gravity and straight-line fits gave errors of 0.809 and 1.219 m. With both frames gravity gave 0.853 m for Regensburg and reached the ground first in the other two cases.

7 Measurement limits and field validation

7.1 Release timing and the interpretation of tracking error

The Regensburg anchor lay about 7 cm horizontally from the spot. The other two anchors were already 0.71 and 1.47 m downfield. None of the three penalties had an independently verified release time within 10 ms or a confirmed direct crossing as its first event. The acceleration test identifies a conflict among the recorded motion, the timing assumption and the launch prior. It cannot distinguish starting the model in mid-flight from timing or tracking error. With just one scoreable late-frame map and no release-synchronized cases, field coverage remains unknown.

A separate short-horizon comparison on ISSIA-3D, a multi-camera reconstruction of one match sequence [9], illustrates what native-time tracking can test. Of 79 continuous segments, 61 supplied two positions 80 ms apart and 47 retained the ball for another 120 ms. On these 47 segments, constant-velocity extrapolation gave a median three-dimensional error of 0.162 m, compared with 0.996 m for holding the last position. The targets were subsequent positions in ordinary play. Penalty validation additionally requires evidence of release and of the first event at or before the goal plane.

7.2 Design of a release-synchronized field test

Synchronized multi-view footage with independent ball-contact annotations would establish release times. Record clock alignment, coordinate calibration, lost frames and intervening contacts. To score a kick, require both early positions by the decision time and an identifiable first event. An interception before the natural crossing should be recorded as a contact or a censored outcome.

Independent calibration sequences can estimate timing bias, per-axis error and cross-frame dependence. A revised model could assign a timing offset to each kick and distinguish a release state from an in-flight anchor. The three existing cases can inform development. Freeze the revised model before testing on held-out matches. Report forecast availability across eligible kicks, point error, coverage and region area on the same entries, and Brier scores on confirmed first events. Reporting area alongside coverage makes the cost of widening a region explicit.

For a goalkeeper, waiting changes both the quality of the forecast and the time available to act. The official median crossing time leaves 299 ms at the original decision and 149 ms after a 250 ms observation. A comparison of a third position, measured spin and measured wind should record both crossing error and remaining action time on the same shots. Estimating save probability would also require the goalkeeper's starting position, movement-onset delay and time-dependent reach.

Conclusion

For the official kick, the most probable goal cell is centred at $(2.562, 2.074)$ m, and the six-cell region in Eq. (5) contains 90.52% of the conditional mass. Two flights within a single force law fit the same early positions and impose a 4.38 cm lower bound on worst-case point error. The compact regions observed in simulated kicks rely on a known release and a specified error model. In professional match data, an acceleration bound shows exactly where those assumptions fail to accommodate the observed motion.

References

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Report on the Use of AI

OpenAI Codex assisted with the physical analysis, numerical implementation, evidence checks, and drafting and review of the English manuscript. This public draft contains the disclosure only. The complete original prompts and outputs are to be appended to the submission copy from the team's interaction records.